Stage 1 Workflow Summary:
From Raw Data to Reference-Site Threshold Selection

1 Overview

This document summarises the three-step workflow that transforms raw sediment-chemistry, environmental, and benthic-invertebrate data into (i) a site-level contamination score, (ii) an ecologically validated scoring rule, and (iii) a defensible reference-site threshold.

2 Step 1: PCA-Based Contamination Scoring

1. **Input.** A 233-site $\times$ 56-variable workbook containing 16 chemical contaminants, 5 environmental variables, and 16 benthic taxa (octave-transformed abundances).
2. **Log-transformation.** Chemical concentrations are transformed via $\log_2(1+x)$ to reduce right skew.
3. **PCA.** Principal Component Analysis is fitted on the 16 log-transformed chemical variables with min–max standardisation. The first five PCs are retained (cumulative variance $\geq 83\%$).
4. **Sign orientation.** Each PC is oriented so that higher scores correspond to higher contamination (positive correlation with known contaminant indicators).
5. **Component rescaling.** Each retained PC score is min–max rescaled to $[0, 1]$.
6. **Two scoring rules.**
   - **SumRel** — sum of rescaled PC scores. Reflects cumulative, multi-dimensional contamination pressure.
   - **MaxRel** — maximum of rescaled PC scores. Captures the single worst contamination axis per site.

Outputs: site-level SumRel and MaxRel scores, PC loadings, variance-explained plot, ridge-loading plot, and corridor-bifurcation maps for each score.

3 Step 2: Ecological Validation via RDA

The competing scores are evaluated by how well the resulting “least-contaminated” reference subset explains taxa composition through environmental gradients.

1. **Reference-site selection.** For a given threshold proportion q (e.g. 25%), the bottom-q fraction of sites ranked by the contamination score is designated as the reference subset.

2. **Redundancy Analysis (RDA).** Within the reference subset, taxa (response, $\mathbf{Y}$) are regressed on z-scored environmental variables (predictors, $\mathbf{X}$). RDA decomposes explained variation into constrained axes (RDA1, RDA2, ...).

3. **Permutation tests.** A global pseudo-F test (999 permutations) assesses overall model significance. Per-axis and per-term tests identify which axes and environmental variables contribute significantly.

4. **Score comparison.** At each tested threshold, SumRel consistently yields higher adjusted R^2 and lower p-values than MaxRel, indicating that the cumulative scoring rule selects reference sites where environmental gradients more cleanly explain biological variation.

4 Step 3: Reference-Threshold Sensitivity Analysis

Rather than fix the threshold at an arbitrary value, a grid sweep from 5% to 99% (step 2%) evaluates how RDA performance changes with reference-subset size.

1. **Metrics collected at each threshold:** R^2, adjusted R^2, global pseudo-F, p-value, constrained and total inertia, RDA1 eigenvalue, and sample size.

2. **Identifying the recommended range.** Adjusted R^2 for SumRel shows a local plateau between approximately 22%–28%, before a steep decline at larger thresholds where contaminated sites dilute the reference pool. Within this range the global test remains highly significant ($p = 0.001$) and sample sizes exceed 50, ensuring statistical power.

3. **F-critical threshold.** On the pseudo-F panel, the F-statistic corresponding to $p = 0.05$ and the threshold at which it is first reached are identified by linear interpolation and reported in the legend.

4. **Selected threshold: 25%.** A 25% cutoff sits in the middle of the recommended plateau, yielding $\sim$58 reference sites with adjusted $R^2 \approx 0.22$ and $p = 0.001$.

5 Final Decisions

- **Scoring rule:** SumRel (ecological view — cumulative contamination pressure).
- **Reference threshold:** 25% of least-contaminated sites.
- **Full RDA** is fitted at the chosen threshold to produce the definitive triplot, axes-summary, and terms-summary tables.

6 Code Organisation

All computation is implemented in the `src/zci` package:

`run_stage1.py`
: Runs PCA + dual scoring (SumRel / MaxRel).

`run_stage_threshold.py`
: Runs Stage 1, sweeps thresholds, fits full RDA at 25%, and saves sensitivity plots, comparison tables, triplot, and RDA summary tables.

The standalone `run_stage_rda.py` is superseded by the integrated threshold-sensitivity pipeline, which performs the same RDA fitting at the selected threshold while also providing the full sensitivity analysis.